

Ideas and structure for the contributed talk.

Biological tissues are highly anisotropic, meaning that their diffusion rates are not the same in every direction. For routine DW imaging we often ignore this complexity and reduce diffusion to a single average value, the apparent diffusion coefficient (ADC), but this is overly simplistic. A superior method to model diffusion in complex materials is to use the diffusion tensor, a 3×3 array of numbers corresponding to diffusion rates in each combination of directions. The three diagonal elements (Dxx, Dyy, Dzz) represent diffusion coefficients measured along each of the principal (x-, y- and z-) laboratory axes. The six off-diagonal terms (Dxy, Dyz, etc) reflect the correlation of random motions between each pair of principal directions.

Coriolis terms

Rescaling.

The Gronwall part.

Are these weak solutions and not L-H WS.

trace and boundary. $h^{-1/2}$

coercivity of the diffusion matrix and the eigenvalue.

why the ϵ^{-3} in the approximated function.

any comment on the lateral conditions?

IMPORTANT: u3 BCs for the limit model case. $un_3 = 0$

weak formulation: the form itself comes from calculations, but the regularities $L^\infty L^2$ and $L^2 H^1$ come simply from definition, not from necessity of the terms, right? For example, L^∞ regularity is not needed explicitly, is it? The regularities for the test function on the other hand come from what is needed for the expressions to make sense.

why do we need $\tilde{u}_H$ to be in H^1 in time? regularities in the weak formulation.

stating the main theorem why can we say that $C_0 = 0$ on the boundary, when it is only an L^2 function on the domain?

stating the theorem — u_0 is there the initial value of u_ϵ, u , or both? — It is the initial value belonging to the anisotropic system. It is needed at

the energy inequality.

the definition of weak solutions: the $\mathbf{u}$ vector is in V . then we conclude at some point, u_3 is in $L^2 L^2 \dots$ but it was by definition in V , which is essentially H^1 . clear this. Proposition1.

Definition3 - $U = (\mathbf{v}, C)$ instead of $\mathbf{v}$ I think it is $\mathbf{u}_H$.